SAMPLE PAPER (made up for testing - not a real exam)

B.Tech Semester V - End Semester Examination May 2023

Natural Language Processing and Generative AI

Time: 3 Hours Max. Marks: 60

Instructions: Attempt all questions. Figures to the right indicate full marks.

Q1. Explain the stages of a typical NLP pipeline with a neat diagram. [10]

Q2. Differentiate between stemming and lemmatization with examples. [5]

Q3. What is a Recurrent Neural Network? Explain the vanishing gradient problem. [10]

Q4. Explain the self-attention mechanism used in the Transformer architecture. [10]

Q5. Write short notes on:

(a) Tokenization [5]

(b) Stop word removal [5]

Q6. What is LangChain? Explain its main components. [10]

Q7. Explain the BLEU score used for evaluating machine translation. [5]